# CompoundIQ — System Architecture Overview

**Version:** 2.0 | **Date:** April 9, 2026 **Status:** Production POC — 19 phases, 86 work orders (81 completed, 5 backlog)

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## 1. Product Overview

CompoundIQ is a B2B2C functional medicine infrastructure platform that connects independent prescribing clinics with compounding pharmacies through intelligent sourcing, automated fulfillment, and transparent pricing.

**The Problem:** Independent clinics prescribe compounded medications at scale but lack infrastructure to do so profitably. Sourcing is manual (phone calls, spreadsheets), margin math is error-prone, and fulfillment is entirely fax-dependent with no tracking or escalation.

**The Solution:** A platform that lets a medical assistant search all licensed pharmacies in seconds, lock a margin, and generate a tokenized SMS checkout link. Upon payment, the system auto-generates a compliant prescription and routes it to the pharmacy via the optimal integration channel.

**Revenue Model:** Per-transaction platform fee captured from the spread between wholesale and retail price. The clinic earns margin on every fill. The pharmacy gets structured, error-free orders. The patient gets seamless guest checkout via SMS.

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## 2. Three Applications, One Codebase

CompoundIQ consists of three distinct web applications within a single Next.js 16 codebase, organized via App Router route groups:

| Application | Route Group | Users | Theme | Viewport |
|---|---|---|---|---|
| **Clinic App** | `/(clinic-app)/` | Medical Assistants, Providers, Clinic Admins | Light mode | Desktop |
| **Ops Dashboard** | `/(ops-dashboard)/` | Internal operations team | Dark mode | Desktop |
| **Patient Checkout** | `/checkout/` | Patients (guest, no login) | Light mode | Mobile-first (320–428px) |

Each application has its own layout, auth guard, error boundaries, and loading states. There is no cross-navigation between applications. Role-based access control is enforced at both the middleware and database level.

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## 3. System Architecture Diagram

```
+-------------------------------------------------------------------+
|                        CLIENT LAYER                               |
|   +----------------+  +----------------+  +-------------------+   |
|   |   Clinic App   |  | Ops Dashboard  |  | Patient Checkout  |   |
|   | (Light Mode)   |  | (Dark Mode)    |  | (Mobile-First)    |   |
|   +-------+--------+  +-------+--------+  +--------+----------+   |
+-----------|-------------------|---------------------|-------------+
            |                   |                     |
```

```
+-----------|-------------------|--------------------|-------------+
|           v                   v                    v             |
|                    VERCEL EDGE NETWORK                            |
|   +-------------------------------------------------------+   |
|   | Edge Middleware: Auth Guards, Rate Limiting, Token Verify  |   |
|   | Global CDN: Static Assets, Page Routes                   |   |
|   +-------------------------------------------------------+   |
+-----------------------------------------------------------------|
            |                   |                    |
+-----------|-------------------|--------------------|-------------+
|           v                   v                    v             |
|                    API LAYER (Serverless)                        |
|   +----------------+  +----------------+  +----------------+      |
|   | Route Handlers |  |Webhook Handlers|  |   Cron Jobs    |      |
|   | (20+ endpoints)|  | (4 sources)    |  | (10 scheduled)  |      |
|   +-------+--------+  +-------+--------+  +-------+--------+      |
+-----------|-------------------|-------------------|---------------+
            |                    |                   |
+-----------|--------------------|-------------------|-------------+
|           v                    v                   v             |
|                   ORDER STATE MACHINE                            |
|   +-------------------------------------------------------+   |
|   | OrderStateMachine: CAS Engine (23 states, 47 transitions) |   |
|   | TransitionValidator | SnapshotManager | WorkflowDispatcher|   |
|   +---------------------------+-----------------------------+   |
+---------------------------------|-------------------------------+
                                  |
            +---------------------+---------------------+
            |                     |                     |
            v                     v                     v
+-------------------+  +-------------------+  +-------------------+
| PHARMACY ADAPTER  |  | PAYMENT ENGINE    |  | SLA ENGINE        |
| 4-Tier Routing    |  | Stripe Connect    |  | 10 SLA Types      |
| Circuit Breaker   |  | Express           |  | 3-Tier Escalation |
| Cascade on Fail   |  | Multi-Party Split |  | 10 Cron Jobs       |
+-------------------+  +-------------------+  +-------------------+
      |                      |                      |
      v                      v                      v
+-------------------+  +-------------------+  +-------------------+
| Tier 1: REST API  |  | Stripe API        |  | Slack Alerts      |
| Tier 2: Playwright|  | (Zero PHI)        |  | PagerDuty Pages   |
| Tier 3: Spec API  |  +-------------------+  | SMS Reminders     |
| Tier 4: Fax/Documo|                         +-------------------+
+-------------------+
            |
+-----------|-------------------------------------------------------+
|           v              DATA LAYER                                |
|   +-------------------------------------------------------+   |
|   | Supabase (PostgreSQL 15+)                                |   |
|   | 33 Tables + 1 View | 10 Enums | Row-Level Security on ALL tables   |   |
|   +-------------------------------------------------------+   |
|   | Supabase Vault    | Supabase Auth      | Supabase Storage |   |
```

```
|   | AES-256-GCM       | JWT Custom Claims  | RLS-Gated Buckets|   |
|   | Pharmacy Creds    | Role-Based Access  | Fax PDFs, Screenshots|
|   +-------------------+--------------------+--------------------+  |
+-----------------------------------------------------------------+
```

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## 4. Technology Stack

| Layer | Technology | Version / Notes |
|---|---|---|
| Frontend | Next.js (App Router) | 16+ with React 19, Server + Client Components |
| UI Framework | Tailwind CSS + Shadcn/UI | Semantic design tokens, dark mode support |
| Backend / Database | Supabase (PostgreSQL) | 15+ with RLS on all 33 tables, Supabase Auth |
| Authentication | Supabase Auth + JWT | Custom claims: app_role, clinic_id in user_metadata |
| Payments | Stripe Connect Express | Multi-party split: patient pays, clinic receives payout, platform captures fee |
| Fax | Documo mFax REST API | HIPAA-compliant, Tier 4 fallback |
| SMS | Twilio Programmable Messaging | Delivery tracking, opt-out handling |
| Portal Automation | Playwright | Headless browser for Tier 2 pharmacy portals |
| Credential Storage | Supabase Vault | AES-256-GCM encryption for all pharmacy credentials |
| Cron Scheduling | Vercel Crons | 10 scheduled jobs defined in vercel.json |
| Deployment | Vercel | Serverless functions, atomic deploys, instant rollback |
| Monitoring | Sentry + Slack + PagerDuty | Error tracking (PHI-scrubbed), ops alerts, critical escalation |

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## 5. The 4-Tier Pharmacy Adapter Layer

The Pharmacy Adapter Layer is the platform's core technical differentiator — a unified integration engine that abstracts pharmacy connectivity behind a single interface, enabling seamless order routing regardless of each pharmacy's technical capability.

| Tier | Method | Pharmacies | Speed | Coverage |
|---|---|---|---|---|
| **Tier 1** | Direct REST API | ReviveRX, Vios, MediVera, LifeFile network (Empower, Belmar, UCP, Strive), Precision | Instant (~2s) | ~25% |
| **Tier 2** | Portal Automation (Playwright) | Olympia/DrScript, Wells/WellsPx3 | ~5 min | ~20% |

| | | | | |
|---|---|---|---|---|
| **Tier 3** | Standardized API Spec | Future partners adopting our published spec | Instant | Future 30%+ |
| **Tier 4** | Fax Fallback (Documo) | Any pharmacy without Tier 1/2/3 capability | ~30 min | Universal |

**Key architectural insight:** LifeFile is the de facto B2B standard for compounding pharmacies. A single LifeFile API integration unlocks Empower, Belmar, UCP, and Strive — the largest pharmacy network in the country.

### Routing Logic

- **Deterministic:** Always selects the highest available tier for each pharmacy
- **Cascade on failure:** Tier 1 fails -> Tier 2 (if available) -> Tier 4 (fax)
- **Circuit breaker:** 3 consecutive failures opens the breaker, 5-minute cooldown, 2 consecutive successes to close (HC-15)
- **Audit trail:** Every submission attempt logged to adapter_submissions table

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## 6. Order Lifecycle — 23-State Machine

Every order flows through a closed state machine with exactly 23 states and 47 valid transitions. All transitions use atomic Compare-And-Swap (CAS) to prevent race conditions.

```
DRAFT
  |
  v
AWAITING_PAYMENT ----> PAYMENT_EXPIRED (72h auto-expiry)
  |
  v
PAID_PROCESSING
  |
  +---> SUBMISSION_PENDING (Tier 1/2/3)
  |       |
  |       +---> PHARMACY_ACKNOWLEDGED ---> PHARMACY_COMPOUNDING
  |       |                                      |
  |       +---> SUBMISSION_FAILED                v
  |               |                       PHARMACY_PROCESSING
  |               +---> REROUTE_PENDING          |
  |                                              v
  +---> FAX_QUEUED (Tier 4)               READY_TO_SHIP
          |                                      |
          +---> FAX_DELIVERED                    v
          |       |                          SHIPPED
          +---> FAX_FAILED                       |
                  |                              v
                  +---> REROUTE_PENDING     DELIVERED (terminal)
```

**Terminal states:** CANCELLED, REFUNDED, DELIVERED **Error states:** ERROR_PAYMENT_FAILED, ERROR_COMPLIANCE_HOLD **Financial recovery:** DISPUTED, REFUND_PENDING

### Snapshot Immutability

When an order transitions from DRAFT to AWAITING_PAYMENT, 7 fields are permanently frozen:

- wholesale_price_snapshot, retail_price_snapshot
- medication_snapshot, shipping_state_snapshot
- provider_npi_snapshot, pharmacy_snapshot
- provider_signature_hash_snapshot

A PostgreSQL trigger ( `prevent_snapshot_mutation()` ) blocks any subsequent UPDATE to these fields. There is no unlock mechanism — to change a locked field, the order must be cancelled and a new one created.

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## 7. Payment Architecture — Stripe Connect Express

```
Patient pays $300 (retail_price)
         |
         v
+------------------+
|  Stripe Connect  |
|   Express        |
+------------------+
         |
    +----+----+
    |         |
    v         v
Platform    Clinic
Fee: $22.50  Payout: $127.50
(15% of     (via transfer_data
 margin)     .destination)
```

- **Single PaymentIntent** per order — never multiple
- **Automatic 3-way split:** Patient pays retail, platform captures fee via `application_fee_amount` , clinic receives payout via `transfer_data.destination`
- **Zero PHI in Stripe:** Metadata contains `order_id` only. Description: "Custom compounded prescription order." No medication names, patient names, or NPI numbers.
- **Stripe processing fees** (2.9% + $0.30) deducted from the platform's portion
- **Transfer failures** do NOT affect order status — they are financial reconciliation events

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## 8. SLA Engine — 10 Enforcement Types

The SLA engine is the platform's time-enforcement backbone. A Vercel Cron job runs every 5 minutes scanning for breached deadlines and triggering automatic escalation.

| SLA Type | Deadline | Trigger |
|---|---|---|
| PAYMENT_EXPIRY | 72 hours | Order enters AWAITING_PAYMENT |
| SMS_REMINDER_24H | 24 hours | Checkout link sent |
| SMS_REMINDER_48H | 48 hours | Checkout link sent (24h before expiry) |
| FAX_DELIVERY | 30 minutes | Fax queued via Documo |
| PHARMACY_ACK | 4 business hours | Fax delivered (Tier 4) |

| | | |
|---|---|---|
| PHARMACY_COMPOUNDING | TAT + 4h buffer | Pharmacy acknowledges |
| SHIPPING | 24 hours | Ready to ship |
| TRACKING_UPDATE | 24 hours | Order shipped |
| ADAPTER_SUBMISSION_ACK | 5–30 min (tier-dependent) | Adapter submission triggered |
| PHARMACY_COMPOUNDING_ACK | 2 business hours | Pharmacy acknowledges (Tier 1/2/3) |

### 3-Tier Escalation

| Tier | Channel | Trigger |
|---|---|---|
| Tier 0 | Automated action (retry, cascade) | At breach |
| Tier 1 | Slack #ops-alerts | At breach (or after Tier 0) |
| Tier 2 | Slack DM to ops lead | 15 min unacknowledged |
| Tier 3 | PagerDuty page | 30 min unacknowledged |

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## 9. Webhook Architecture — 4 External Sources

All webhooks follow a 7-step processing pipeline: Receive -> Authenticate -> Extract event_id -> Idempotency check -> Process -> Record outcome -> Respond 200.

| Source | Events | Auth Method |
|---|---|---|
| **Stripe** | payment_intent.succeeded, charge.dispute.created, transfer.failed, account.updated | HMAC-SHA256 (whsec_*), 5-min timestamp tolerance |
| **Documo** | fax.delivered, fax.failed, fax.received (inbound) | Token-based verification |
| **Twilio** | message.status (delivered, failed, undelivered) | X-Twilio-Signature HMAC-SHA1 |
| **Pharmacy APIs** | order.acknowledged, order.compounding, order.rejected, order.shipped | Per-pharmacy HMAC-SHA256 via Supabase Vault |

**Critical invariant:** The system ALWAYS returns HTTP 200 to webhook senders — even on internal failures. This prevents retry storms. Failures are handled via the internal Dead Letter Queue.

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## 10. Database Schema

### 33 Tables with Full Row-Level Security

**Core Entities (12 tables):** clinics, providers, patients, pharmacies, pharmacy_state_licenses, catalog, catalog_history, orders, order_status_history, webhook_events, order_sla_deadlines, inbound_fax_queue

**Pharmacy Adapter Layer (5 tables):** pharmacy_api_configs, pharmacy_portal_configs, adapter_submissions, normalized_catalog, pharmacy_webhook_events

**Hierarchical Medication Catalog (8 tables — WO-82):** ingredients, salt_forms, dosage_forms, routes_of_administration, formulations, formulation_ingredients, pharmacy_formulations, sig_templates

**Provider Speed Features (3 tables — WO-85):** provider_favorites, protocol_templates, protocol_items

**Regulatory Compliance (5 tables — WO-86):** epcs_audit_log, drug_interactions, patient_protocol_phases, phase_advancement_history + TOTP columns on providers (totp_secret_encrypted, totp_enabled, totp_verified_at)

**Views:** provider_prescribing_history (aggregates order data for adaptive shortlist)

### 10 Enum Types

order_status_enum (23 values), stripe_connect_status_enum (4), cancellation_reason_enum (5), regulatory_status_enum (3), webhook_source_enum (4), sla_type_enum (10), inbound_fax_status_enum (5), integration_tier_enum (4), adapter_submission_status_enum (8), catalog_source_enum (3)

### Key Schema Principles

- All monetary values: NUMERIC(10,2) — never floating point
- Soft-delete on all tables (deleted_at + is_active) — no physical DELETEs
- Append-only audit tables for all state transitions (order_status_history, epcs_audit_log, phase_advancement_history)
- TOTP secrets encrypted with AES-256-GCM at rest
- Supabase Vault for all pharmacy credentials (UUID references, never plaintext)
- Foreign keys enforced everywhere
- Hierarchical catalog: ingredients → salt_forms → formulations → pharmacy_formulations

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## 11. Infrastructure — 10 Vercel Cron Jobs

| Endpoint | Schedule | Purpose |
|---|---|---|
| /api/cron/sla-check | Every 5 min | SLA breach scanning and escalation |
| /api/cron/sla-refire | Every 5 min | Re-escalate stalled SLA deadlines |
| /api/cron/payment-expiry | Every 15 min | Auto-expire unpaid orders at 72 hours |
| /api/cron/adapter-health-check | Every 10 min | Circuit breaker state updates |
| /api/cron/submission-reconciliation | Every 30 min | Detect stuck submissions |
| /api/cron/portal-status-poll | Every 30 min | Tier 2 portal status polling |
| /api/cron/fax-retry | Every 5 min | Fax retry with exponential backoff |
| /api/cron/screenshot-cleanup | Every hour | Remove expired Tier 2 screenshots |
| /api/cron/daily-digest | 14:00 UTC daily | Ops pipeline health summary to Slack |
| /api/cron/poc-credential-sync | 05:00 UTC daily | Reset POC demo accounts to canonical passwords |

All cron endpoints are protected by CRON_SECRET bearer token authentication.

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## 12. Security & HIPAA Compliance

### Data Protection

| Control | Implementation |
|---|---|
| Data at rest | AES-256 encryption (Supabase-managed) |
| Data in transit | TLS 1.2+ on all connections |
| Credential storage | Supabase Vault (AES-256-GCM) — never plaintext |
| Access control | Row-Level Security on all 33 tables |
| EPCS 2FA | TOTP authenticator (DEA 21 CFR 1311) for controlled substances |
| TOTP secrets | AES-256-GCM encrypted at rest (not plaintext) |
| Session management | 30-min idle timeout, 12-hour absolute max |
| Audit trail | Append-only tables for all state transitions + EPCS audit log |
| Drug interactions | Known interaction pairs checked at review with severity alerts |
| PHI in Stripe | ZERO — metadata contains order_id only |
| PHI in SMS | Minimal — first name + URL only, never medication names |
| PHI in monitoring | Sentry beforeSend scrubs all PHI before transmission |
| Supabase Realtime | DISABLED — hard HIPAA requirement (leaks PHI via WebSocket) |

### Authentication Model

| Context | Method | Access Level |
|---|---|---|
| Clinic users | Supabase Auth JWT (app_role + clinic_id) | Own clinic data only |
| Ops admins | Supabase Auth JWT (app_role = ops_admin) | Cross-clinic read access |
| Patients | Stateless JWT in URL (72h expiry, HS256) | Single order only |
| Webhooks | Service-specific signature verification | Service role (bypasses RLS) |
| Cron jobs | CRON_SECRET bearer token | Service role |

### Role-Based Access Control (4 roles)

| Role | Clinic App | Ops Dashboard | Patient Checkout |
|---|---|---|---|
| clinic_admin | Full access (own clinic) | Blocked | N/A |
| provider | Signature + review only | Blocked | N/A |
| medical_assistant | Full workflow (own clinic) | Blocked | N/A |
| ops_admin | Blocked | Full access (all clinics) | N/A |
| patient (anon) | Blocked | Blocked | Single order via JWT |

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## 13. Key Metrics

| Metric | Value |
|---|---|
| PostgreSQL tables | 33 |
| Enum types | 10 |
| Order statuses | 23 (with 47 valid transitions) |
| Hard constraints | 16 (HC-01 through HC-16) |
| SLA types | 10 |
| Integration tiers | 4 |
| Cron jobs | 10 |
| Build phases completed | 19 |
| Work orders completed | 81 (5 backlog) |
| Applications | 3 |
| User roles | 4 (+ patient anonymous) |
| Webhook sources | 4 (Stripe, Documo, Twilio, Pharmacy APIs) |
| Drug interaction pairs | 6 (seeded, expandable) |
| Provider favorites | Per-provider saved prescription configs |
| Protocol templates | Per-clinic multi-medication bundles |
| Environment variables | 40+ across 8 categories |

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## 14. Deployment Architecture

```
GitHub (master branch)
         |
         v
    Vercel CI/CD
    (Automatic on push)
         |
    +----+----+
    |         |
    v         v
Production   Preview
(atomic      (per PR,
 deploy)      staging
              creds)
```

- **Git-driven:** Every push to master auto-deploys to production
- **Atomic deploys:** Old version serves traffic until new version is fully ready

- **Instant rollback:** One-click restore to any prior deployment
- **Preview environments:** Every PR gets an isolated preview URL with staging credentials
- **Zero-downtime migrations:** 5-phase pattern (additive -> dual-write -> backfill -> tighten -> cleanup)

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*Built with Next.js 16, Supabase, Stripe Connect Express, Twilio, Documo mFax, and Playwright. Deployed on Vercel serverless infrastructure. Designed for HIPAA-adjacent compliance with zero PHI exposure to non-covered entities.*